

How Private Is the Cut? Measuring Image and Label Leakage in Split Federated Learning

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Abstract—Split federated learning (SFL) trains a shared model across clients without moving their raw data: the network is cut in two, each client runs the first few layers, and only the intermediate activations (the “smashed data”) travel to the server. That design is sold as privacy-preserving. This report measures how much privacy it actually provides, on one common testbed: a CIFAR-adapted ResNet18 on CIFAR-10, four cut depths, three seeds. A decoder trained by an honest-but-curious server reconstructs client images from the smashed data almost perfectly at a shallow cut (41.8 dB PSNR, 0.995 SSIM at cut 1) and degrades steadily with depth (24.3 dB, 0.81 at cut 4), confirming prior results. The backward channel behaves differently: a linear probe on the gradient returned across the cut recovers 10-class labels at 56% accuracy at cut 1, rising to 69–72% at deeper cuts, against a 10% chance rate. Image leakage falls with depth while label leakage rises, so no single cut position is safest for both. Finally, Gaussian noise on the smashed data (the PixelDP-style defense the SplitFed paper itself proposes) is swept along the privacy–utility frontier: accuracy holds near 88% up to $\sigma = 4$ while reconstruction SSIM falls from 0.98 to 0.85, but even $\sigma = 8$ only degrades the attack (21.7 dB, 0.74 SSIM) at a cost of six accuracy points. The individual attacks reproduce published findings; the contribution here is putting both leak directions and a defense on the same SFL setup and making the opposite-direction effect explicit. Code and results: <https://github.com/s6hib/splitfed-privacy>.

I. INTRODUCTION

Machine learning models are trained on data that is often private: medical records, or just someone’s photo roll. Federated learning (FL) keeps that data on the device by having each client train a full local model and share only weight updates [1]. Split learning (SL) goes after a different constraint, client compute: the network is cut at a chosen layer, the client runs only the layers below the cut, and sends the activations at the cut to a server that runs the rest [2], [3]. Those activations are called smashed data. SplitFed learning (SFL) combines the two, training split clients in parallel and averaging their model halves with FedAvg [4].

In all of these, the raw input never leaves the client, which is why the family gets pitched as a privacy-preserving alternative to centralized training. The promise turns out to be shakier than it sounds. The smashed data is a deterministic function of the input, computed by a few convolutional layers, and a server that wants the input back can try to learn the inverse map. The backward direction leaks too: in a two-party split where one side holds the labels, the gradient that flows back across the cut is shaped by the loss and therefore by the label itself [5].

The first phase of this project [6] reproduced the five training methods from the SplitFed paper (centralized, FL, SL, SFLV1, SFLV2) on CIFAR-10 with a CIFAR-adapted ResNet18, fixed three bugs in the reference implementation, and made the cut position configurable. It found that deeper cuts help SFLV1 on both accuracy and communication. This second phase asks the natural follow-up: what does the cut position do to *privacy*? Three experiments run on the same testbed, numbered steps 2–4 to match the project’s phases and the released code (step 1 is the training harness):

- **Feature inversion (forward channel).** An honest-but-curious server trains a decoder to reconstruct client images from smashed data, at each of four cut depths.
- **Label leakage (backward channel).** In a two-party split where labels stay with the label party, two attacks try to recover labels from the gradient returned at the cut.
- **A noise defense.** Gaussian noise on the smashed data, swept over σ to trace the privacy–utility frontier at a fixed cut.

A literature check (Section II) shows that each attack, taken alone, reproduces a published result. This report is therefore framed as a reproduction-and-measurement study, not a discovery. What it adds is the combination: both leak directions and a defense, measured on one SFL setup with multiple seeds. That combination makes an observation explicit that the literature leaves scattered across papers: image leakage and label leakage move in *opposite* directions as the cut deepens. “Cut deeper for more privacy” is good advice against inversion and bad advice against label leakage.

A. Threat model

All attacks here are passive (honest-but-curious): the attacker follows the protocol and never corrupts training; it only computes on what the protocol already shows it.

For feature inversion, the attacker is a server that holds the trained client-side encoder and data from the same distribution as the clients’. In SFL this is realistic: the fed server receives every client’s encoder weights to average them, so any honest-but-curious party on the server side sees exactly this model. The attacker pushes its own images through the encoder to build (activation, image) pairs, trains a decoder on them, and applies the decoder to clients’ smashed data.

For label leakage, the setting is the two-party split of Li et al. [5]: the non-label party holds the bottom model and sends activations up; the label party computes the loss and returns the gradient with respect to the cut activations. The non-label

party is the attacker and sees only that returned gradient. The unsupervised attack uses gradient norms alone. The supervised probe also assumes the attacker knows labels for some of its own examples, and measures how much label information is linearly decodable from the gradient, an upper bound rather than a guaranteed attack.

II. RELATED WORK

SplitFed. Thapa et al. [4] define the two SFL variants used here: SFLV1 averages both client-side and server-side models with FedAvg; SFLV2 keeps a single shared server model. The same paper already proposes the defenses this project measures (client-side differential privacy and PixelDP-style noise on the smashed data), so Section IV-C is a measurement of their proposal, not a new defense.

Inverting intermediate activations. He et al. [7] is the canonical version of the Step-2 attack: train a decoder from activations back to inputs in collaborative inference, sweep the split point, and report PSNR/SSIM per depth, with recovery quality dropping as the split deepens. They treat SSIM below roughly 0.3 as unrecognizable. Lee et al. [8] run the same style of attack inside SFL itself and find SSIM consistently decreasing with deeper cuts, on Fashion-MNIST with a small CNN, single seed, no defenses. FORA [9] builds a stronger feature-oriented reconstruction attack on ResNet18/CIFAR-10 across all split points and benchmarks it against defenses in plain split learning. SDAR [10] reconstructs features and infers labels with a passive honest-but-curious attacker that uses adversarial regularization, and analyzes how attack quality degrades with split depth. UnSplit [11] and PCAT [12] push the attacker further (recovering inputs with no auxiliary data at all, or stealing the model’s functionality outright through a pseudo-client), again across split points. The depth-versus-reconstruction curve in this report is a confirmatory reproduction of this line of work, on the SFLV1 topology.

Label leakage from gradients. Li et al. [5] show that in two-party split learning the gradient returned across the cut leaks the label, starting from the observation that gradient norms separate the classes in binary tasks, and propose protections. Step 3 adapts their norm attack to CIFAR-10 by pairing it with a binary vehicle-versus-animal split, and adds a supervised linear probe for the 10-class case, swept across cut depth.

Defenses in split settings. ResSFL [13] addresses model inversion in SFL with a learned resistant encoder and reports the same accuracy-versus-reconstruction frontier shape that Step 4 traces; their defense is a trained bottleneck rather than noise. The noise defense measured here is the simpler one already present in the SplitFed paper [4].

Individually, then, each curve in this report exists somewhere in the literature. The gap this project fills is small and practical: one testbed (ResNet18, CIFAR-10, SFLV1 topology, seeds held in common) on which the forward-channel attack, the backward-channel attack, and a defense are all run, so their trends can be compared directly rather than across papers with different models, datasets, and training budgets.

III. EXPERIMENTAL SETUP

Testbed. All experiments use CIFAR-10 [14] (50,000 train / 10,000 test, 32×32 RGB) and a CIFAR-adapted ResNet18 [15]: 3×3 stem with stride 1, no initial max-pool, adaptive average pooling, about 11.2M parameters at the standard width. The model splits into a client part and a server part at one of four cut positions: cut 1 is after the stem; cuts 2–4 move the boundary after residual layer groups 1, 2, and 3. The smashed data shapes are (64, 32, 32) at cuts 1–2, (128, 16, 16) at cut 3, and (256, 8, 8) at cut 4. The training harness, bug fixes, and cut parameterization come from phase one [6].

Step 1: training the targets. SFLV1 is trained at each cut with 5 clients, IID data, Adam at learning rate 10^{-4} , batch size 256, one local epoch per round, 50 rounds, for seeds {42, 1234, 2026}, and the converged client encoders are checkpointed. Final-round test accuracy is $74.5 \pm 1.4\%$ (cut 1), $76.5 \pm 0.7\%$ (cut 2), $76.6 \pm 0.4\%$ (cut 3), and $78.9 \pm 0.7\%$ (cut 4) (mean $\pm$ std over seeds), consistent with phase one’s single-seed finding that deeper cuts help SFLV1 [6]. The full five-method utility comparison (centralized 86.7%, SL 85.0%, SFLV2 83.2%, FL 81.1%, SFLV1 77.0% at cut 2, 50 epochs, seed 1234) was run in phase one and is not repeated here; this phase trains only what the attacks need.

Step 2: feature inversion. For each (cut, seed), the attacker trains a decoder from the frozen, trained encoder’s outputs back to images in $[0, 1]$. The decoder is a stack of nearest-neighbor upsampling and convolution blocks (width 128) ending in a sigmoid; it trains on the CIFAR-10 training images pushed through the encoder, with MSE loss, Adam at 10^{-3} , 30 epochs. Reconstruction quality is measured on the 10,000 test images: PSNR computed per-image then averaged, and SSIM. Each seed’s attack targets that seed’s own trained encoder.

Step 3: label leakage. For each (cut, seed), a plain two-party split model trains for 15 epochs (Adam, 10^{-3} , batch 256), reaching 85–89% test accuracy. Per-sample gradients at the cut are then captured for 4,000 test images, in evaluation mode so batch normalization uses running statistics and one sample’s gradient does not depend on its batchmates. Two attacks run on the captured gradients. The norm attack [5] scores each sample by the ℓ_2 norm of its cut gradient and is evaluated as AUC on the binary vehicles-versus-animals split of CIFAR-10’s classes, folded to $[0.5, 1]$ since the attacker can flip its decision. The probe attack fits a linear softmax classifier (gradient $\rightarrow$ class) on 70% of the captured gradients, standardized per-dimension, and reports top-1 accuracy on the held-out 30%; chance is 10%.

Step 4: noise defense. Following the PixelDP-style defense in [4], Gaussian noise $\mathcal{N}(0, \sigma^2)$ is added to the smashed data before it crosses the cut, during both training and inference, with fresh noise every forward pass. At cut 2 and seed 1234, a two-party split model trains for 25 epochs at each $\sigma \in \{0, 0.1, 0.25, 0.5, 1, 2, 4, 8\}$; test accuracy is measured with the noise active (as deployed), and the Step-2 decoder is then retrained for 25 epochs against the frozen noisy encoder

TABLE I

STEP 2: RECONSTRUCTION QUALITY VS. CUT DEPTH (MEAN $\pm$ STD, 3 SEEDS).

Cut	PSNR (dB)	SSIM	MSE
1	41.83 $\pm$ 1.82	0.995 $\pm$ 0.001	0.00008
2	35.71 $\pm$ 0.41	0.983 $\pm$ 0.001	0.00030
3	29.81 $\pm$ 0.15	0.938 $\pm$ 0.001	0.00117
4	24.30 $\pm$ 0.27	0.809 $\pm$ 0.007	0.00415

to measure what an attacker still recovers. $\sigma = 0$ is the undefended reference. This sweep is single-seed; its trend, not its third digit, is the result.

IV. RESULTS

A. Step 2: image reconstruction falls with depth

Table I and Fig. 1 give the inversion results. Reconstruction quality falls monotonically as the cut deepens, from 41.8 dB PSNR / 0.995 SSIM at cut 1 to 24.3 dB / 0.81 at cut 4, with tight error bars across seeds. The direction and shape match He et al. [7] and Lee et al. [8].

Two things stand out. First, a cut-1 reconstruction at 0.995 SSIM is, for practical purposes, the original image: the first convolutional block preserves nearly everything. Second, even the deepest cut here does not reach safety. He et al. treat SSIM below about 0.3 as unrecognizable [7]; cut 4 sits at 0.81, and the reconstruction grids in Fig. 2 stay clearly recognizable. Within this architecture’s available cuts, depth shrinks the leak but never closes it.

B. Step 3: label leakage rises with depth

Table II and Fig. 3 give the label-leakage results. The unsupervised norm attack leaks at every depth, with AUC flat around 0.64–0.65 against the 0.5 no-leak line. The supervised probe recovers 10-class labels at $55.9 \pm 3.0\%$ accuracy at cut 1 and jumps to roughly 69–72% at cuts 2–4, plateauing after cut 2. Chance is 10%; the trained models themselves score 85–89%, so the gradient alone carries a large fraction of what the full model knows about the label.

The comparison with Step 2 is the point. Moving the cut from 1 to 4 drives image leakage down (SSIM 0.995 $\rightarrow$ 0.81) and label leakage up (probe 56% $\rightarrow$ 72%). A practitioner choosing the cut position is not turning a privacy dial; they are trading one leak for the other. The likely mechanism is unsurprising: deeper activations are more abstract and harder to invert pixel-by-pixel, but the gradients at deeper layers sit closer to the loss and are more aligned with class identity. The data here shows the two opposite trends on one testbed; pinning down the mechanism is left as future work.

C. Step 4: the noise defense degrades, but does not stop, inversion

Table III and Fig. 4 trace the frontier at cut 2. Small noise is nearly free for both sides: at $\sigma \leq 0.25$, accuracy stays within one to two points of the 90.1% undefended baseline and the attacker’s PSNR barely moves. From $\sigma = 0.5$ to $\sigma = 4$

TABLE II

STEP 3: LABEL LEAKAGE FROM THE CUT GRADIENT (MEAN $\pm$ STD, 3 SEEDS). CHANCE ACCURACY IS 10%; NO-LEAK AUC IS 0.5.

Cut	Norm-leak AUC	Probe accuracy (%)
1	0.651 $\pm$ 0.032	55.9 $\pm$ 3.0
2	0.637 $\pm$ 0.032	69.0 $\pm$ 2.9
3	0.643 $\pm$ 0.018	71.7 $\pm$ 2.0
4	0.637 $\pm$ 0.041	71.9 $\pm$ 2.4

TABLE III

STEP 4: PRIVACY–UTILITY FRONTIER UNDER SMASHED-DATA NOISE (CUT 2, SEED 1234).

σ	Accuracy (%)	PSNR (dB)	SSIM
0 (undefended)	90.1	34.3	0.983
0.1	88.1	34.3	0.982
0.25	89.0	34.2	0.978
0.5	88.8	31.8	0.968
1	87.9	30.2	0.952
2	88.0	28.1	0.919
4	88.3	24.9	0.849
8	84.1	21.7	0.742

the defense earns its keep: reconstruction drops from 31.8 to 24.9 dB (SSIM 0.97 $\rightarrow$ 0.85) while accuracy holds near 88%, because training with the noise in the loop lets the model adapt to it. At $\sigma = 8$ the model finally pays (84.1%), and the attack is degraded but alive: 21.7 dB, SSIM 0.74, still above the recognizability threshold and visibly so.

For comparison, $\sigma = 8$ of noise at cut 2 buys slightly more reconstruction degradation than moving the cut to 4 does (21.7 vs. 24.3 dB), and both leave recognizable images. The honest summary: this defense bends the frontier; it does not break it.

V. DISCUSSION

There is no single “more private” cut. The two attack surfaces respond to depth in opposite directions. Against inversion, every extra layer between the input and the cut destroys pixel information, and the measurements show the standard steady decline. Against label leakage, depth helps the attacker: by cut 2 the returned gradient already supports nearly 70% 10-class label recovery. Deployment advice that only considers one channel will get the other one wrong. A system that must protect both needs either a defense on each channel or a cut chosen for the threat that actually matters in its setting.

The measured defense is weak and not free. Noise on the smashed data is attractive because it directly corrupts what the inversion attacker reads, and it does bend the curve. But over this sweep it never pushes reconstructions below recognizability, and the strongest setting costs six accuracy points. These results agree with the broader pattern in the defense literature, where simple noise is a baseline that stronger learned defenses (e.g., resistant encoders [13]) are built to beat.

Limitations. Step 4 is single-seed, so only its trend should be read. Step 3’s probe plateaus after cut 2 rather than rising strictly. The two attacks also run on different training

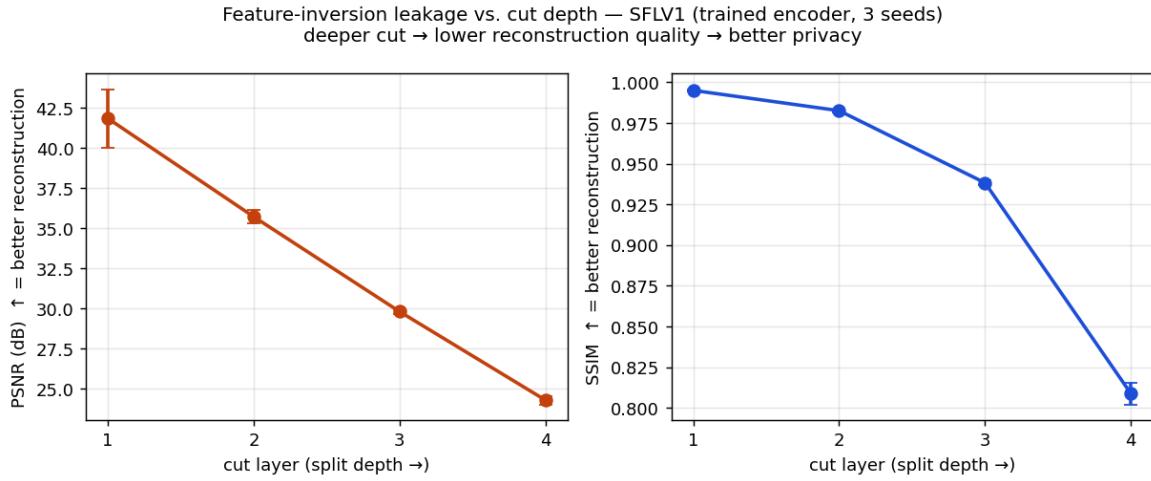


Fig. 1. Feature-inversion leakage vs. cut depth on trained SFLV1 encoders (3 seeds, mean $\pm$ std). Reconstruction quality falls steadily as the split moves deeper.



Fig. 2. Originals (top row) and reconstructions (bottom row) at cut 1 (upper panel) and cut 4 (lower panel), seed 1234. Cut-1 reconstructions are visually indistinguishable from the originals; cut-4 reconstructions are blurrier but still clearly recognizable.

topologies, each matching its standard threat model (inversion attacks the federated SFLV1 encoders, label leakage uses the two-party split of [5]), so cross-step comparisons are qualitative. The inversion attack targets the final converged encoder; attacking the moving, repeatedly-averaged encoder during training is untested here. Everything is CIFAR-10 and IID, on one architecture. And since the supervised probe assumes the attacker has some labeled examples of its own, its numbers are a ceiling on what leaks, not an attack every non-label party could mount.

Future work. DP-SGD is the natural companion defense, but it perturbs parameter gradients during training rather than the forward activations, so it should do little against inversion; whether it blunts gradient label leakage is the interesting measurement, and it was not built here. A direct SFLV1-versus-SFLV2 privacy comparison on this testbed would close a gap the literature also leaves open. Non-IID partitions and the label protections proposed by Li et al. [5] round out the list.

VI. CONCLUSION

On a single SplitFed testbed, this report measured both directions of the privacy question that the cut layer is supposed to answer. The forward channel leaks images, badly at shallow cuts and recognizably even at the deepest cut available. The backward channel leaks labels, and it leaks more once the cut moves past the shallowest position, the opposite trend. A Gaussian-noise defense on the smashed data shifts the privacy–utility frontier but leaves reconstructions recognizable at every accuracy-tolerable setting. The individual curves confirm published results; together they make the practical point that in split learning, the cut position is a trade between leak surfaces, not a privacy knob. Code, results, and figures are at <https://github.com/s6hib/splitfed-privacy>.

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Step 3 — gradient label leakage vs cut depth (splitnn, 3 seeds)

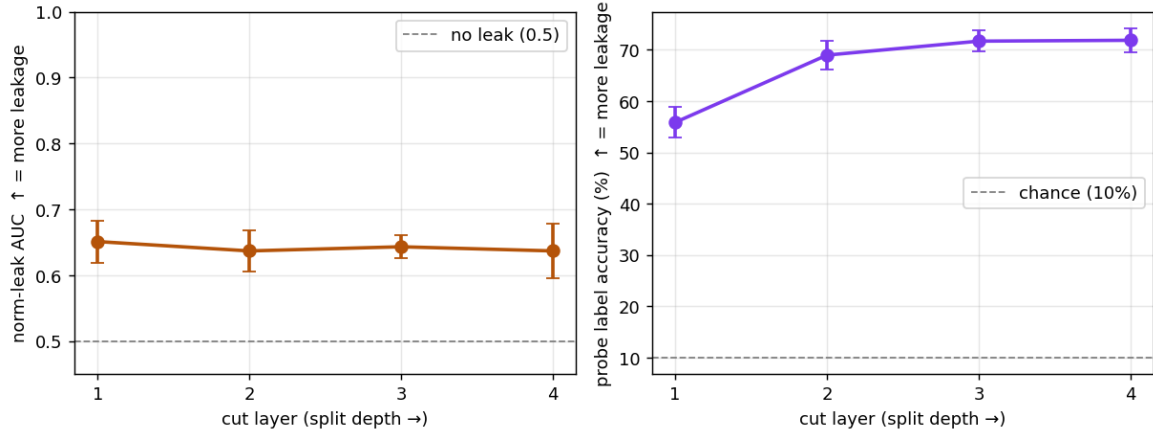


Fig. 3. Gradient label leakage vs. cut depth in two-party split learning (3 seeds). Left: unsupervised norm-leak AUC stays near 0.64 at every depth. Right: the supervised probe rises from 56% at cut 1 to about 72% at deeper cuts, the opposite direction from image leakage.

Step 4 — privacy-utility frontier under smashed-data noise (PixelDP)
each point a noise level σ ; up-right = leaky, down-left = private

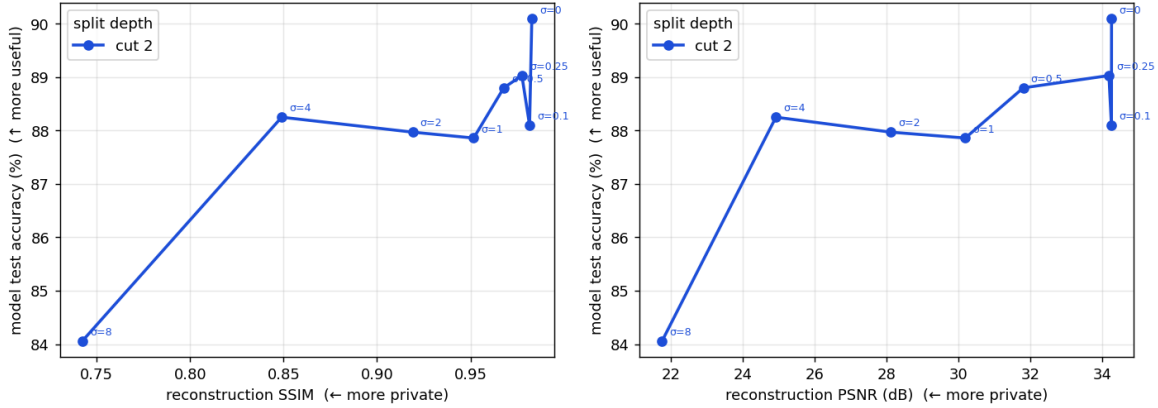


Fig. 4. Privacy-utility frontier under Gaussian noise on the smashed data (cut 2, seed 1234). Each point is one noise level σ ; the undefended point anchors the top right. Accuracy holds near 88% out to $\sigma = 4$ while reconstruction quality falls, but even $\sigma = 8$ leaves the attack degraded rather than defeated.

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